2024, Mar. 2024 Release LOC:Acute Adult
Hematology/Oncology: Treatments

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction:
For patients who are experiencing adverse event(s) related to an outpatient treatment or a complication related to their disease progression from a hematologic or oncologic issue, please see the Hematology or Oncology: Complications or Disease Progression subset.
Introduction:
This subset is for patients who need admission for work up and treatment for new diagnosis of acute leukemia or lymphoma, chemotherapy initiation, bone marrow or stem cell transplantation, protocol-driven CAR T-cell therapy induction, or inpatient radiation therapy.

Evaluation and Treatment:
Patients who are being evaluated for acute Leukemia (e.g., acute myeloid leukemia, acute lymphoblastic leukemia) or lymphoma require inpatient admission and prompt treatment based on labs (e.g., complete blood count, blood smear) and biopsy results. A diagnosis can be confirmed with a bone marrow, lymph node, or tissue biopsy.
Chemotherapy is a form of pharmacotherapy that is used in the systemic treatment of cancer. Many chemotherapy agents can be safely and effectively administered in the outpatient setting. However, this subset is intended for the administration of chemotherapy agents that require an extended period for time that preclude the agent from being administered in the outpatient setting. This subset is also intended for those treatments that need to be administered in the inpatient setting because the patient is at high risk for an adverse event (e.g., tumor lysis syndrome).
CAR T-cell (chimeric antigen receptor) therapy is a type of immunotherapy that uses immune cells called T-cells that are genetically altered in a lab to enable them in locating and destroying cancer cells more effectively. CAR T-cell therapy is approved to treat several types of hematological malignancies including Leukemia, lymphoma and multiple myeloma.
Radiation therapy can frequently be provided in the outpatient setting. However, radiation to treat brain malignancies, brain metastasis, spinal cord compression, or solid tumors that have caused SVC (superior vena cava) syndrome frequently require close inpatient monitoring and intervention. Radioactive iodine ablations and radioactive implants also require inpatient monitoring along with extended periods of isolation.
This subset includes patients requiring bone marrow or stem cell transplantation who need inpatient admission through the continuum for pre-treatment with myeloablative or sub-myeloablative therapy, transplantation, and post-treatment engraftment.
This subset is appropriate for patients who have a primary brain malignancy, CNS metastasis, or spinal cord compression requiring inpatient radiation or chemotherapy for treatment.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, **One:**

- **Episode Day 1, One:**
  - (Symptom or finding within 24h)
  - **ACUTE, One:** (1)
    - Acute leukemia or lymphoma suspected and, **Both:**
      - Finding, ≥ **Two:**
        - Hepatosplenomegaly or lymphadenopathy
        - Mediastinal mass
        - Platelet count < 100,000/cu.mm($100x10^9$/L)
        - Temperature ≥ 100.4°F(38.0°C) (2)
        - WBC, ≥ **One:**
          - > 50,000/cu.mm($50x10^9$/L)
          - Blasts or myelocytes
          - Lymphocytes > 100,000/cu.mm($100x10^9$/L)
      - Bone marrow, lymph node, or tissue biopsy scheduled or performed within 24h (3)
    - Chemotherapy, ≥ **One:** (4, 5)
      - Risk for adverse effect, ≥ **One:** (6)
        - Single or multidrug regimen requiring continuous infusion > 16h (7)
        - Single or multidrug regimen, ≥ **One:**
          - Elranatamab, Talquetamab-tgvs or Teclistamab-cqyv (8)
          - EPOCH, DA-EPOCH or R-EPOCH
          - High dose ifosfamide
          - Hyper-CVAD
        - Finding, ≥ **One:**
          - Acute lymphoblastic leukemia
          - Acute myeloid leukemia and, ≥ **One:**
            - WBC ≥ 25,000/cu.mm($25x10^9$/L)
            - Lactate dehydrogenase (LDH) > 2x ULN
          - Chronic lymphocytic leukemia and, ≥ **One:**
            - Fludarabine
            - Rituximab
            - WBC ≥ 50,000/cu.mm($50x10^9$/L)
          - Hematologic malignancy and abnormal renal function (9)
          - Non-Hodgkin's lymphoma, **One:**
            - Burkitt
            - Intermediate grade non-Hodgkin's lymphoma and lactate dehydrogenase (LDH) > 2x ULN
            - Lymphoblastic
          - Solid tumor, bulky or advanced stage
        - Urotoxic agent administration and, **Both:** (10)
          - IV fluid (11)
          - Urine specific gravity or pH monitoring
      - High dose interleukin-2 ≤ 5d, **All:** (12)
        - Dose > 600,000 IU/kg
        - Continuous cardiac monitoring (excludes Holter)
        - IV fluid (11)
      - High dose methotrexate, **All:** (13)
        - Dose > 500 mg/m^2
        - IV fluid (11)
        - Sodium bicarbonate (includes PO)
    - Pre-bone marrow or stem cell transplant ≤ 10d, **One:** (14, 15)
      - Inpatient sub-myeloablative therapy
      - Inpatient myeloablative therapy

- Pre-CAR T-cell therapy administration, ≤ 6d [16]
- ⊖ Radiation and, ≥ **One:**
  - Brain malignancy, brain metastasis or spinal cord compression
  - Iodine ablation > 30 mCi [17]
  - Radioactive implant
  - SVC syndrome confirmed by imaging
- ⊖ **INTERMEDIATE, ≥ One:** [18]
  - Bone marrow or stem cell transplant performed within 24h [14]
  - CAR T-cell therapy administration performed within 24h
  - HFNC and respiratory monitoring every 3-4h [19, 20]
  - NIPPV [21]
  - Oxygen ≥ 40%(0.40) ≤ 2d [22]
- ⊖ **CRITICAL, ≥ One:** [23]
  - CRRT [24]
  - ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** [25]
    - MAP < 65 mmHg
    - Systolic blood pressure < 90 mmHg and < baseline
  - Hemodynamic monitoring, invasive [26]
  - HFNC and respiratory monitoring every 1-2h [19, 20, 22]
  - Mechanical ventilation
  - ⊖ NIPPV and, ≥ **One:** [21]
    - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
    - ⊖ Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
      - COPD and arterial or venous pH ≤ 7.24
      - No hx of COPD
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- ⊖ **Episode Day 2, One:**
  - ⊖ **ACUTE, One:**
    - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [27]
      - Afebrile
      - ⊖ Treatment, ≥ **One:**
        - ⊖ Chemotherapy regimen, ≥ **One:**
          - Completed
          - Discontinued
          - Manageable in post-acute setting
        - Acute leukemia or lymphoma ruled out or treatment not indicated
        - CAR T-cell therapy completed or not indicated
        - Bone marrow transplant or stem cell transplant completed or not indicated
        - ⊖ Radiation, **Both:**
          - Isolation discontinued
          - No exposure risk to others
      - Lab values within acceptable limits [28]
      - Tolerating PO or nutritional route established [29, 30]
      - ⊖ Complication or comorbidity, ≥ **One:**
        - No complication or active comorbidity relevant to this episode of care [31]
        - No evidence of cytokine release syndrome or neurotoxicity [32]
    - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - ⊖ AKI and, **Both:** [33]
        - ⊖ Finding, ≥ **One:**
          - Urine output < 0.5 mL/kg/h
          - ⊖ Creatinine, **One:**
            - ≥ 1.5x baseline and chronic kidney disease

            - ≥ 1.5x ULN [34]
        - GFR > 25%(0.25) decrease from baseline [35]
        - Dialysis initiated
    - Intervention, ≥ **One:**
        - Volume expander ≤ 2d [36, 37]
        - Diuretic ≥ 2x/24h, ≤ 3d
        - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [38]
            - Diuretic (includes PO)
            - Nephrotoxic agent (includes PO) [39]
        - Dialysis (initial) ≤ 7d since initiation [40]
        - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [41]
- Acute leukemia or lymphoma suspected and biopsy pending ≤ 6d
- Chemotherapy, infusion or post infusion and, ≥ **One:** [4, 5]
    - Blood product transfusion
    - Interleukin-2 ≤ 5d and, **Both:** [12]
        - Continuous cardiac monitoring (excludes Holter)
        - IV fluid [11]
    - Methotrexate post infusion monitoring and, **All:** [13]
        - Level ≥ 0.05 μM/L
        - IV fluid [11]
        - Methotrexate toxicity requiring folinic acid (includes PO) or glucarpidase
        - Sodium bicarbonate (includes PO)
    - Single or multidrug regimen requiring continuous infusion > 16h [7]
    - Single or multidrug regimen, ≥ **One:**
        - Elranatamab, Talquetamab-tgvs or Teclistamab-cqyv [8]
        - EPOCH, DA-EPOCH or R-EPOCH
        - High dose ifosfamide
        - Hyper-CVAD
    - Tumor lysis syndrome prophylaxis, **All:** [42]
        - Laboratory monitoring at least daily
        - IV fluid [11]
        - Medication, **One:** [43]
            - Allopurinol (includes PO)
            - Rasburicase, administered or post administration
    - Urotoxic agent administered or ≤ 24h since infusion completed and, **Both:**
        - IV fluid [11]
        - Urine specific gravity or pH monitoring
- Fever, new onset and, **Both:** [44]
    - Temperature ≥ 100.4°F(38.0°C) [2]
    - Intervention, ≥ **One:**
        - Culture pending ≤ 2d
        - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d [45]
        - Imaging study ≤ 24h
- Hyperviscosity syndrome and, **Both:** [46]
    - Finding, ≥ **One:**
        - Mucosal bleeding
        - Neurological symptoms
        - Visual changes
    - Intervention, **All:**
        - Apheresis [47]
        - Hydration or nutrition, ≥ **One:**
            - IV fluid [11, 48]
            - Advancing diet as tolerated

- Neurological assessment at least 3x/24h [49]
- ⊖ Inadequate oral intake ≤ 7d and, ≥ **One:** [50]
  - ⊖ Antiemetic, ≥ **One:**
    - Antiemetic ≥ 3x/24h
    - Dexamethasone [51]
    - Serotonin antagonist at least daily
  - IV fluid [11, 48]
  - Parenteral nutrition
  - Tube feeding
- ⊖ Intractable diarrhea or colitis and, ≥ **One:** [52, 53]
  - Corticosteroid [54]
  - Immunosuppressant therapy (includes PO and SC) [55]
  - IV fluid bolus [48]
- ⊖ Intractable pain, ≥ **One:**
  - Conversion from IV to PO or transdermal ≤ 24h
  - ⊖ Analgesic ≤ 2d, ≥ **One:** [56]
    - ≥ 3x/24h
    - Continuous
- ⊖ Jaundice or hyperbilirubinemia and, **Both:** [57]
  - ⊖ Finding, ≥ **One:**
    - PT ≥ 1.5x ULN
    - INR ≥ 2.0
    - Total bilirubin ≥ 2x ULN
  - ⊖ Intervention, ≥ **One:**
    - ERCP scheduled or performed within last 24h
    - Fresh frozen plasma (FFP)
    - Liver biopsy
    - MRCP scheduled or performed within last 24h
    - Percutaneous drainage performed within last 24h
- ⊖ Pre-bone marrow or stem cell transplant ≤ 10d, **One:** [14, 15]
  - Inpatient sub-myeloablative therapy
  - Inpatient myeloablative therapy
- Pre-CAR T-cell therapy administration, ≤ 6d [16]
- ⊖ Radiation ≤ 7d and, ≥ **One:**
  - Brain malignancy, brain metastasis or spinal cord compression
  - Iodine ablation > 30 mCi [17]
  - Radioactive implant
  - SVC syndrome confirmed by imaging
- ⊖ Thrombocytopenia (excludes known Von Willebrand disease) and, **Both:** [58, 59]
  - ⊖ Finding, ≥ **One:**
    - Platelet count < 20,000/cu.mm($20x10^9$/L)
    - Purpura or petechiae, progressive
  - ⊖ Intervention, ≥ **One:**
    - Blood product transfusion
    - Corticosteroid
    - Immunoglobulin [60]
- ⊖ Tumor lysis syndrome and, **Both:** [42]
  - ⊖ Finding, ≥ **Two:**
    - Calcium < 7.0 mg/dL(1.8 mmol/L)
    - Potassium > 6.0 mg/L(6.0 mmol/L) and no ECG changes
    - Phosphate > 4.5 mg/dL(1.5 mmol/L)
    - Uric acid > 8.0 mg/dL(475.9 µmol/L)
  - ⊖ Intervention, **One:**

- ⊖ Dialysis not required, **Both:**
  - − IV fluid [11, 61]
  - ⊖ Medication, **One:** [43]
    - − Allopurinol (includes PO)
    - − Rasburicase
- − Hemodialysis ≤ 7d since initiation
- − Post Critical care ≤ 24h

⊖ **INTERMEDIATE,** ≥ **One:** [18]
- ⊖ Arrhythmia, ≥ **One:**
  - − IV antiarrhythmic and hemodynamically stable
  - ⊖ Antiarrhythmic conversion from IV to PO, **One:**
    - − No proarrhythmic risk ≤ 24h
    - − Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [62]
- − Bone marrow or stem cell transplant performed within 24h [14, 63]
- − CAR T-cell therapy administration performed within 24h
- − Post-CAR T-cell therapy administration, ≤ 30d
- ⊖ Cytokine release syndrome, actual or suspected and, **Both:** [64]
  - ⊖ Temperature ≥ 100.4°F(38.0°C) and, ≥ **One:** [2]
    - − O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [35]
    - − Systolic BP 90-110 mmHg and unresponsive to IV fluids
  - ⊖ Intervention, ≥ **One:**
    - − Anticytokine therapy
    - − Corticosteroid (includes PO)
    - − IV fluid [11, 48]
- ⊖ Neurotoxicity, actual or suspected, **Both:**
  - ⊖ Finding, ≥ **One:**
    - − Aphasia [65]
    - − Apraxia
    - − Difficulty counting numbers backward
    - − Disorientation to time or place
    - − New or worsening tremor
    - − Unable to write a sentence
  - ⊖ Neurological assessment every 3-4h and, ≥ **One:**
    - − IV fluid [11, 48]
    - − Corticosteroid (includes PO)
- − HFNC and respiratory monitoring every 3-4h [19, 20]
- ⊖ Hyperkalemia, **All:** [66, 67]
  - − Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
  - ⊖ ECG changes, ≥ **One:** [67]
    - − Peaked T waves
    - − PVCs > 6/min
  - ⊖ Intervention, ≥ **One:**
    - − Calcium chloride
    - − Calcium gluconate
    - − Glucose 50%(0.50) with insulin [68]
- ⊖ Post-bone marrow or stem cell transplant and, **All:**
  - − Awaiting engraftment
  - − Isolation
  - ⊖ Hydration or nutrition, ≥ **One:**
    - − IV fluid [11, 48]
    - − Advancing diet as tolerated or nutritional route established [29]
- − NIPPV [21]
- − Oxygen ≥ 40%(0.40) ≤ 2d [22]

- ⊖ Seizure and, **Both:**
  - Activity resolved
  - Anticonvulsant continuous
- ⊖ **CRITICAL,** ≥ **One:** (23)
  - ⊖ Anticonvulsant and, ≥ **One:**
    - Recurrent seizure (69)
    - Refractory seizure (69)
  - ⊖ Arrhythmia, ≥ **One:**
    - Urgent electrical cardioversion performed within last 24h
    - Defibrillation performed within last 24h
    - Temporary pacemaker insertion performed within last 2d
    - ⊖ Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, ≥ **One:** (70)
      - Chest pain
      - ⊖ Heart failure and, ≥ **One:**
        - Arterial Po_2 < 56 mmHg(7.4 kPa)
        - O_2 sat < 89%(0.89) and < baseline
      - ⊖ Hemodynamic instability, ≥ **One:**
        - Heart rate > 120/min, sustained (71)
        - ⊖ Systolic blood pressure, ≥ **One:**
          - < 90 mmHg
          - < 110 mmHg with chronic hypertension
          - > 30 mmHg decrease from baseline (35)
          - Labile (72)
        - MAP < 65 mmHg
  - ⊖ Cerebral edema and, ≥ **One:**
    - Corticosteroid
    - ICP monitoring (includes OR placement)
    - Neurological assessment every 1-2h (49)
    - ⊖ Hyperosmolar therapy, ≥ **One:** (73)
      - Mannitol
      - Hypertonic saline ≥ 3%(0.03) solution
  - Coma, stupor, obtundation, or GCS ≤ 8 (49, 74)
  - CRRT (24)
  - DIC and blood product transfusion (75)
  - ⊖ GVHD, Grade IV and, ≥ **One:** (76)
    - Apheresis (47)
    - Immunosuppressant (77)
  - ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** (25)
    - MAP < 65 mmHg
    - Systolic blood pressure < 90 mmHg and < baseline
  - Hemodynamic monitoring, invasive (26)
  - HFNC and respiratory monitoring every 1-2h (19, 20, 22)
  - Mechanical ventilation
  - ⊖ NIPPV and, ≥ **One:** (21)
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h (78)
  - ⊖ Tumor lysis syndrome and, **Both:**
    - ⊖ Finding, ≥ **One:**
      - Calcium < 5.0 mg/dL(1.25 mmol/L) (79, 80)
      - ⊖ Hyperkalemia, **Both:**
        - Potassium > 6.0 mEq/L(6.0 mmol/L)
        - ⊖ Finding, ≥ **One:**

                - Loss of P wave
                - Multifocal PVCs
                - Neuromuscular deficit [81]
                - Widening QRS
            - Sodium, **One:**
                - < 120 mEq/L(120 mmol/L) and volume overload
                - > 160 mEq/L(160 mmol/L)
        - Intervention, ≥ **One:**
            - Calcium repletion
            - IV fluid resuscitation ≤ 2d [82]
            - Diuretic, continuous
            - Dialysis, **One:**
                - Hemodialysis ≤ 7d since initiation
                - Peritoneal dialysis ≤ 7d since initiation
            - Glucose 50%(0.50) with insulin [68]
    - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3-X, One:**
    - **ACUTE, One:**
        - **Responder**, discharge expected today if clinically stable last 12h, **All:** [27]
            - Afebrile
            - Anti-infective, **One:**
                - Arranged
                - Completed
                - Not indicated
            - Treatment, ≥ **One:**
                - Chemotherapy regimen, ≥ **One:**
                    - Completed
                    - Discontinued
                    - Manageable in post-acute setting
                - Acute leukemia or lymphoma ruled out or treatment not indicated
                - CAR T-cell therapy completed or not indicated
                - Bone marrow transplant or stem cell transplant completed or not indicated
                - Radiation, **Both:**
                    - Isolation discontinued
                    - No exposure risk to others
            - Medication regimen established and tolerated
            - Mental status or GCS within normal limits or returned to baseline [35, 74]
            - Lab values within acceptable limits [28]
            - Tolerating PO or nutritional route established [29, 30]
            - Vital signs stable or within acceptable limits [28]
            - Complication or comorbidity, **One:**
                - No complication or active comorbidity relevant to this episode of care [31]
                - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
                    - AKI and, ≥ **One:**
                        - Kidney function improving or within acceptable limits [28]
                        - Outpatient dialysis regimen established
                    - COPD exacerbation resolving
                    - Decrease in frequency and severity of diarrhea
                    - Dyspnea resolved and, ≥ **One:**
                        - Home oxygen arranged
                        - O_2 sat within acceptable limits [28]
                    - Functional impairment and, **One:** [83]
                        - Discharge to home and, **One:** [84]

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services
- GVHD resolving or stabilized
- Heart failure resolving and, **Both:**
  - Dyspnea resolving or at baseline [35]
  - O_2 sat ≥ 92%(0.92) or within acceptable limits [28]
- Hyperviscosity syndrome resolving and, **Both:**
  - Neurological stability [85]
  - No evidence of bleeding
- Jaundice improved or stabilized
- No evidence of cytokine release syndrome or neurotoxicity [32]
- No seizure activity or baseline [35]
- Pain controlled or manageable [86]
- Spinal cord compression resolving and neurological stability [85]
- Tumor lysis syndrome resolved or stabilized

- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - AKI and, **Both:** [33]
    - Finding, ≥ **One:**
      - Urine output < 0.5 mL/kg/h
      - Creatinine, **One:**
        - ≥ 1.5x baseline and chronic kidney disease
        - ≥ 1.5x ULN [34]
      - GFR > 25%(0.25) decrease from baseline [35]
      - Dialysis initiated
    - Intervention, ≥ **One:**
      - Volume expander ≤ 2d [36, 37]
      - Diuretic ≥ 2x/24h, ≤ 3d
      - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [38]
        - Diuretic (includes PO)
        - Nephrotoxic agent (includes PO) [39]
      - Dialysis (initial) and ≤ 7d since initiation [40]
      - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [41]
  - Acute leukemia or lymphoma suspected and biopsy pending ≤ 6d
  - Anti-infective and, ≥ **One:**
    - Temperature ≥ 100.4°F(38.0°C) [2]
    - ANC < 500/cu.mm($500x10^6$/L) [87]
    - WBC ≥ 12,000/cu.mm($12x10^9$/L)
    - Bands > 10%(0.10)
  - Chemotherapy, infusion or post infusion and, ≥ **One:** [5]
    - Blood product transfusion
    - Interleukin-2 ≤ 5d and, **Both:** [12]
      - Continuous cardiac monitoring (excludes Holter)
      - IV fluid [11]
    - Methotrexate post infusion monitoring and, **All:** [13]
      - Level ≥ 0.05 µM/L
      - IV fluid [11]
      - Methotrexate toxicity requiring folinic acid (includes PO) or glucarpidase
      - Sodium bicarbonate (includes PO)
    - Single or multidrug regimen requiring continuous infusion > 16h [7]
    - Single or multidrug regimen, ≥ **One:**
      - Elranatamab, Talquetamab-tgvs or Teclistamab-cqyv [8]
      - EPOCH, DA-EPOCH or R-EPOCH

            - High dose ifosfamide
            - Hyper-CVAD
        - Tumor lysis syndrome prophylaxis, **All:** [42]
            - Laboratory monitoring at least daily
            - IV fluid [11]
            - Medication, **One:** [43]
                - Allopurinol (includes PO)
                - Rasburicase, administered or post administration
        - Urotoxic agent administered or ≤ 24h since infusion completed and, **Both:**
            - IV fluid [11]
            - Urine specific gravity or pH monitoring
    - COPD and, **All:**
        - Finding, ≥ **One:**
            - O_2 sat ≤ 91%(0.91) and < baseline [35]
            - Arterial Po_2 ≤ 60 mmHg(8.0 kPa) [88]
            - Arterial or venous Pco_2 > baseline [35, 88]
        - Bronchodilator ≥ 4x/24h [89]
        - Corticosteroid (includes PO or inhaled)
        - Oxygenation, **One:**
            - O_2 sat ≥ 92%(0.92) or baseline [35]
            - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [22, 35]
    - GVHD and immunosuppressant and, ≥ **One:** [76, 77, 90]
        - Conversion from IV to PO ≤ 2d [91]
        - Medication adjustment (includes PO) ≤ 3d [38, 92]
        - New medication initiation
    - Fever, new onset and, **Both:** [44]
        - Temperature ≥ 100.4°F(38.0°C) [2]
        - Intervention, ≥ **One:**
            - Culture pending ≤ 2d
            - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d [45]
            - Imaging study ≤ 24h
    - Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ **One:** [83, 93]
        - Cognitive evaluation and retraining
        - OT evaluation and training
        - PT evaluation and training
        - Speech-language evaluation and training
        - Swallowing evaluation and retraining
    - Heart failure and, **Both:** [94]
        - Dyspnea > baseline and, ≥ **One:** [35]
            - O_2 sat ≤ 91%(0.91) and < baseline [35]
            - Worsening edema
            - Heart failure confirmed by imaging [95]
        - Intervention, ≥ **One:** [96]
            - Diuretic, **One:** [97]
                - ≥ 2x/24h (includes PO)
                - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** [38]
                    - Blood urea nitrogen (BUN) increased from baseline [35]
                    - Creatinine increased from baseline [35]
            - Dialysis
    - Hyperviscosity syndrome ≤ 6d since onset and, **Both:** [98]
        - Finding, ≥ **One:**
            - Mucosal bleeding
            - Persistent neurological symptoms

- − Visual changes
- ⊖ Intervention, ≥ **One:**
  - − Apheresis (47)
  - ⊖ Hydration or nutrition, ≥ **One:**
    - − IV fluid (11, 48)
    - − Advancing diet as tolerated
  - − Neurological assessment at least 3x/24h, ≤ 2d (49)
- ⊖ Inadequate oral intake ≤ 7d and, ≥ **One:** (50)
  - ⊖ Antiemetic, ≥ **One:**
    - − Antiemetic ≥ 3x/24h
    - − Dexamethasone (51)
    - − Serotonin antagonist at least daily
  - − IV fluid (11, 48)
  - − Parenteral nutrition
  - − Tube feeding
- ⊖ Intractable diarrhea or colitis and, ≥ **One:** (52, 53)
  - − Corticosteroid (54)
  - − Immunosuppressant therapy (includes PO and SC) (55)
  - − IV fluid bolus (48)
- ⊖ Intractable pain, ≥ **One:**
  - − Conversion from IV to PO or transdermal ≤ 24h
  - ⊖ Analgesic ≤ 2d, ≥ **One:** (56)
    - − ≥ 3x/24h
    - − Continuous
- ⊖ Jaundice or hyperbilirubinemia and, ≥ **One:**
  - − ERCP scheduled or performed within last 24h
  - ⊖ Fresh frozen plasma (FFP) and, ≥ **One:**
    - − PT ≥ 1.5x ULN
    - − INR ≥ 2.0
  - − MRCP scheduled or performed within last 24h
  - − Liver biopsy performed within last 24h
  - − Percutaneous drain requiring repositioning within last 24h
  - − Percutaneous drainage performed within last 24h
- ⊖ Neutropenia, **All:** (99, 100)
  - ⊖ ANC, **One:** (87)
    - − < 500/cu.mm(500x10^6/L)
    - − < 1000/cu.mm(1000x10^6/L) with documented decline (101)
  - − Anti-infective
  - − Temperature ≥ 100.4°F(38.0°C) (2)
- ⊖ Pneumonia confirmed by imaging and, **Both:** (102)
  - ⊖ Finding, ≥ **One:**
    - − Temperature ≥ 100.4°F(38.0°C) (2)
    - − Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 (74, 103)
    - − O_2 sat ≤ 91%(0.91) and < baseline (35)
    - − WBC ≥ 12,000/cu.mm(12x10^9/L)
    - − WBC ≤ 4,000/cu.mm(4 x10^9/L)
    - − Bands > 10%(0.10)
    - − ANC < 500/cu.mm(500x10^6/L) (87)
  - ⊖ Intervention, ≥ **One:** (104)
    - − Requiring supplemental oxygen (22)
    - − Anti-infective (includes PO)
    - − Respiratory interventions at least 6x/24h (78)
- ⊖ Pre-bone marrow or stem cell transplant ≤ 10d, **One:** (14, 15)

- Inpatient sub-myeloablative therapy
- Inpatient myeloablative therapy
- Pre-CAR T-cell therapy administration, ≤ 6d
- Post-bone marrow or stem cell transplant and, ≥ **One:**
  - Post-engraftment, **Both:**
    - Finding, ≥ **One:**
      - ANC < 500/cu.mm(500x10^6/L)
      - Platelet count ≤ 20,000/cu.mm(20x10^9/L)
    - Hydration or nutrition, ≥ **One:**
      - IV fluid [11, 48]
      - Advancing diet as tolerated or nutritional route established [29]
  - GVHD and immunosuppressant and, ≥ **One:** [76, 77, 90]
    - Conversion from IV to PO ≤ 2d [91]
    - Medication adjustment (includes PO) ≤ 3d [38, 92, 105]
    - New medication initiation ≤ 2d
- Radiation ≤ 7d and, ≥ **One:**
  - Brain malignancy, brain metastasis or spinal cord compression
  - Iodine ablation > 30 mCi [17]
  - Radioactive implant
  - SVC syndrome confirmed by imaging
- Thrombocytopenia ≤ 7d since onset and, **Both:**
  - Finding, ≥ **One:**
    - Bleeding actual or suspected and, ≥ **One:** [106]
      - Hct < 25%(0.25) or Hb < 8.3 g/dL(83 g/L)
      - Platelet count < 60,000/cu.mm(60x10^9/L)
      - Heart rate 100-120/min, sustained [71]
      - Hct or Hb decreased last 24h
      - Orthostatic hypotension, **Both:** [107]
        - Sustained drop in blood pressure within 3 min of sitting or standing [71]
        - Blood pressure, ≥ **One:**
          - Systolic drop ≥ 20 mmHg
          - Diastolic drop ≥ 10 mmHg
      - PT ≥ 1.5x upper limit normal (ULN) or INR ≥ 2.0
      - PTT ≥ 1.5x upper limit normal (ULN)
    - Platelet count < 50,000/cu.mm(50x10^9/L)
    - Purpura or petechiae, progressive or unknown etiology
  - Intervention, ≥ **One:**
    - Blood product transfusion
    - Corticosteroid (includes PO)
    - Immunoglobulin [60]
- Tumor lysis syndrome and, ≥ **One:**
  - Dialysis not required, **Both:**
    - IV fluid [11, 61]
    - Medication, **One:** [43]
      - Allopurinol (includes PO)
      - Rasburicase
  - Hemodialysis ≤ 7d since initiation
- Post Critical care ≤ 24h

Non-responder, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

INTERMEDIATE, ≥ One: [18]

- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Arrhythmia, ≥ **One:**
    - IV antiarrhythmic and hemodynamically stable
    - Antiarrhythmic conversion from IV to PO, **One:**
      - No proarrhythmic risk ≤ 24h
      - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [62]
  - Bone marrow or stem cell transplant performed within 24h [14, 63]
  - Post-CAR T-cell therapy administration, ≤ 30d
  - Cytokine release syndrome and, ≥ **One:** [64]
    - Anticytokine therapy
    - Corticosteroid (includes PO)
    - IV fluid [11, 48]
  - Neurotoxicity, actual or suspected, **Both:**
    - Finding, ≥ **One:**
      - Aphasia [65]
      - Apraxia
      - Difficulty counting numbers backward
      - Disorientation to time or place
      - New or worsening tremor
      - Unable to write a sentence
    - Neurological assessment every 3-4h and, ≥ **One:**
      - IV fluid [11, 48]
      - Corticosteroid (includes PO)
  - GVHD and, **One:** [76]
    - Grade I or II and monoclonal antibody initiation ≤ 3 doses [108]
    - Grade III and immunosuppressant [77]
  - HFNC and respiratory monitoring every 3-4h [19, 20]
  - Hyperkalemia, **All:** [66, 67]
    - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
    - ECG changes, ≥ **One:** [67]
      - Peaked T waves
      - PVCs > 6/min
    - Intervention, ≥ **One:**
      - Calcium chloride
      - Calcium gluconate
      - Glucose 50%(0.50) with insulin [68]
  - Post-bone marrow or stem cell transplant and, **All:**
    - Awaiting engraftment
    - Isolation
    - Hydration or nutrition, ≥ **One:**
      - IV fluid [11, 48]
      - Advancing diet as tolerated or nutritional route established [29]
  - NIPPV [21]
  - Oxygen ≥ 40%(0.40) ≤ 2d [22]
  - Seizure and, **Both:**
    - Activity resolved
    - Anticonvulsant continuous
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review

- **CRITICAL, ≥ One:** [23]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

- ⊖ Anticonvulsant and, ≥ **One:**
  - − Recurrent seizure [69]
  - − Refractory seizure [69]
- ⊖ Arrhythmia, ≥ **One:**
  - − Urgent electrical cardioversion performed within last 24h
  - − Defibrillation performed within last 24h
  - − Temporary pacemaker insertion performed within last 2d
  - ⊖ Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, ≥ **One:** [70]
    - − Chest pain
    - ⊖ Heart failure and, ≥ **One:**
      - − Arterial Po_2 < 56 mmHg(7.4 kPa)
      - − O_2 sat < 89%(0.89) and < baseline
    - ⊖ Hemodynamic instability, ≥ **One:**
      - − Heart rate > 120/min, sustained [71]
      - ⊖ Systolic blood pressure, ≥ **One:**
        - − < 90 mmHg
        - − < 110 mmHg with chronic hypertension
        - − > 30 mmHg decrease from baseline [35]
        - − Labile [72]
      - − MAP < 65 mmHg
- ⊖ Cerebral edema and, ≥ **One:**
  - − Corticosteroid
  - − ICP monitoring (includes OR placement)
  - − Neurological assessment every 1-2h [49]
  - ⊖ Hyperosmolar therapy, ≥ **One:** [73]
    - − Mannitol
    - − Hypertonic saline ≥ 3%(0.03) solution
- − Coma, stupor, obtundation or GCS ≤ 8 [49, 74]
- − CRRT [24]
- − DIC and blood product transfusion [75]
- ⊖ GVHD, Grade IV and, ≥ **One:** [76]
  - − Apheresis [47]
  - − Immunosuppressant [77]
- ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** [25]
  - − MAP < 65 mmHg
  - − Systolic blood pressure < 90 mmHg and < baseline
- − Hemodynamic monitoring, invasive [26]
- − HFNC and respiratory monitoring every 1-2h [19, 20, 22]
- − Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [21]
  - − FiO_2 > 50%(0.50)
  - − Respiratory intervention every 1-2h [78]
- ⊖ Tumor lysis syndrome and, **Both:**
  - ⊖ Finding, ≥ **One:**
    - − Calcium < 5.0 mg/dL(1.25 mmol/L) [79, 80]
    - ⊖ Hyperkalemia, **Both:**
      - − Potassium > 6.0 mEq/L(6.0 mmol/L)
      - ⊖ Finding, ≥ **One:**
        - − Loss of P wave
        - − Multifocal PVCs
        - − Neuromuscular deficit [81]
        - − Widening QRS

- Sodium, **One:**
  - < 120 mEq/L(120 mmol/L) and volume overload
  - > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One:**
  - Calcium repletion
  - IV fluid resuscitation ≤ 2d [82]
  - Diuretic, continuous
  - Dialysis, **One:**
    - Hemodialysis ≤ 7d since initiation
    - Peritoneal dialysis ≤ 7d since initiation
  - Glucose 50%(0.50) with insulin [68]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review

Discharge, Level of Care, One: [109]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [110]
    - Patient or caregiver demonstrates ability to manage care
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [111]
    - Medication reconciliation [112]
    - Follow-up care planned [113]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [110]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [114, 115]
    - Skilled services, ≥ **One:** [116]
      - Skilled nursing [117]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [118]
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [111]
    - Medication reconciliation [112]
    - Follow-up care planned [113]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [119]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [120]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [121]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h (117)
- ⊖ Therapy, **All:**
  - Goal directed therapy with rehabilitation potential (122, 123)
  - Functional impairments requiring at least supervision (124)
  - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- ⊖ Complete prior to facility transfer, **All:**
  - Medication reconciliation (112)
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation
- ⊖ **SUBACUTE MEDICAL OR THERAPY, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h (117)
      - ⊖ Therapy, **All:**
        - Goal directed therapy with rehabilitation potential (122, 123)
        - ≥ 2 functional impairments requiring at least minimum assistance (124)
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation (112)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **SUBACUTE REHAB, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, **All:**
      - Goal directed therapy with rehabilitation potential (122, 123)
      - ≥ 2 functional impairments requiring at least minimum assistance (124)
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation (112)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **ACUTE REHAB, Both:**
  - ⊖ Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - ⊖ Services, **All:**
      - Goal directed therapy with rehabilitation potential (122, 123)
      - ≥ 2 functional impairments requiring at least minimum assistance (124)
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines (125)
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation (112)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **LONG TERM ACUTE CARE, Both:**
  - ⊖ Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level

- In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
  - Primary condition or illness and treatment of active comorbid condition(s) [126]
  - Ventilator dependent ≥ 6h/d and weaning planned [127, 128]
  - Acute and comorbid conditions requiring prolonged hospitalization [129]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [112]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
018 CHIMERIC ANTIGEN RECEPTOR (CAR) T-CELL AND OTHER IMMUNOTHERAPIES	**n/a**	**12.9**	**CMS GMLOS**
054 NERVOUS SYSTEM NEOPLASMS WITH MCC	**n/a**	**4**	**CMS GMLOS**
055 NERVOUS SYSTEM NEOPLASMS WITHOUT MCC	**n/a**	**3**	**CMS GMLOS**
124 OTHER DISORDERS OF THE EYE WITH MCC OR THROMBOLYTIC AGENT	**n/a**	**3.4**	**CMS GMLOS**
125 OTHER DISORDERS OF THE EYE WITHOUT MCC	**n/a**	**2.3**	**CMS GMLOS**
146 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITH MCC	**n/a**	**5.5**	**CMS GMLOS**
147 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITH CC	**n/a**	**3.6**	**CMS GMLOS**
148 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITHOUT CC/MCC	**n/a**	**2.2**	**CMS GMLOS**
180 RESPIRATORY NEOPLASMS WITH MCC	**n/a**	**4.9**	**CMS GMLOS**
181 RESPIRATORY NEOPLASMS WITH CC	**n/a**	**3.2**	**CMS GMLOS**
182 RESPIRATORY NEOPLASMS WITHOUT CC/MCC	**n/a**	**2.2**	**CMS GMLOS**
314 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH MCC	**n/a**	**5.1**	**CMS GMLOS**
315 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH CC	**n/a**	**2.7**	**CMS GMLOS**
316 OTHER CIRCULATORY SYSTEM DIAGNOSES WITHOUT CC/MCC	**n/a**	**1.8**	**CMS GMLOS**
374 DIGESTIVE MALIGNANCY WITH MCC	**n/a**	**5.6**	**CMS GMLOS**
375 DIGESTIVE MALIGNANCY WITH CC	**n/a**	**3.6**	**CMS GMLOS**
376 DIGESTIVE MALIGNANCY WITHOUT CC/MCC	**n/a**	**2.4**	**CMS GMLOS**

Condition or Procedure	% Pd Obs	LOS (days)	Type
435 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITH MCC	n/a	4.8	CMS GMLOS
436 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITH CC	n/a	3.4	CMS GMLOS
437 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
542 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITH MCC	n/a	5.2	CMS GMLOS
543 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITH CC	n/a	3.5	CMS GMLOS
544 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
595 MAJOR SKIN DISORDERS WITH MCC	n/a	5.7	CMS GMLOS
596 MAJOR SKIN DISORDERS WITHOUT MCC	n/a	3.5	CMS GMLOS
597 MALIGNANT BREAST DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
598 MALIGNANT BREAST DISORDERS WITH CC	n/a	3.4	CMS GMLOS
599 MALIGNANT BREAST DISORDERS WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
606 MINOR SKIN DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
607 MINOR SKIN DISORDERS WITHOUT MCC	n/a	3	CMS GMLOS
643 ENDOCRINE DISORDERS WITH MCC	n/a	5.1	CMS GMLOS
644 ENDOCRINE DISORDERS WITH CC	n/a	3.6	CMS GMLOS
645 ENDOCRINE DISORDERS WITHOUT CC/MCC	n/a	2.7	CMS GMLOS
686 KIDNEY AND URINARY TRACT NEOPLASMS WITH MCC	n/a	5.3	CMS GMLOS
687 KIDNEY AND URINARY TRACT NEOPLASMS WITH CC	n/a	3.3	CMS GMLOS
688 KIDNEY AND URINARY TRACT NEOPLASMS WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
722 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5.5	CMS GMLOS
723 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.5	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
724 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2	CMS GMLOS
754 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5.1	CMS GMLOS
755 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.3	CMS GMLOS
756 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
834 ACUTE LEUKEMIA WITHOUT MAJOR O.R. PROCEDURES WITH MCC	n/a	9.9	CMS GMLOS
835 ACUTE LEUKEMIA WITHOUT MAJOR O.R. PROCEDURES WITH CC	n/a	4.6	CMS GMLOS
836 ACUTE LEUKEMIA WITHOUT MAJOR O.R. PROCEDURES WITHOUT CC/MCC	n/a	2.7	CMS GMLOS
840 LYMPHOMA AND NON-ACUTE LEUKEMIA WITH MCC	n/a	6.8	CMS GMLOS
841 LYMPHOMA AND NON-ACUTE LEUKEMIA WITH CC	n/a	4	CMS GMLOS
842 LYMPHOMA AND NON-ACUTE LEUKEMIA WITHOUT CC/MCC	n/a	2.7	CMS GMLOS
843 OTHER MYELOPROLIFERATIVE DISORDERS OR POORLY DIFFERENTIATED NEOPLASTIC DIAGNOSES WITH MCC	n/a	5.4	CMS GMLOS
844 OTHER MYELOPROLIFERATIVE DISORDERS OR POORLY DIFFERENTIATED NEOPLASTIC DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS
845 OTHER MYELOPROLIFERATIVE DISORDERS OR POORLY DIFFERENTIATED NEOPLASTIC DIAGNOSES WITHOUT CC/MCC	n/a	2.7	CMS GMLOS
ACUTE LEUKEMIA	12%	14.6	InterQual
LYMPHOMA	16%	8.7	InterQual
NON-HODGKIN’S LYMPHOMA	18%	5.7	InterQual
SVC Syndrome	21%	3.9	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

2:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

3:

Acute leukemia or lymphoma may be confirmed by a bone marrow, lymph node or tissue biopsy (e.g., biopsy of stomach for gastric lymphoma).

4:

Chemotherapy includes:

- Induction chemotherapy
- Reinduction chemotherapy
- Salvage chemotherapy

5:

Reinduction or salvage chemotherapy refers to the treatment of relapsed or refractory acute leukemia following initial induction chemotherapy.

6:

In the setting of chemotherapy administration, factors contributing to the risk of adverse effects include the potential for acute toxicity (e.g., renal failure, hemorrhagic cystitis, uncontrolled vomiting, abnormal blood counts), as well as the administration of complex regimens increasing the susceptibility for complications. Inpatient hospitalization may be required to ensure patient safety through prolonged observation, frequent monitoring, and/or institution of prevention measures.

7:

Instruction: For this criterion, continuous chemotherapy is an infusion (including IV fluids as part of the regimen) that is given continuously for more than 16 hours. Secondary medical review may be required when outpatient facilities cannot provide care for 16 hours. This criterion is intended to be used for chemotherapy drugs given over a long period of time.

8:

The initiation of the immunotherapy agents indicated for relapsed or refractory multiple myeloma require a step-up dosing regimen to closely monitor for systemic signs or symptoms related to cytokine release syndrome (CRS) and neurotoxicity including the goal of reducing the incidence and severity of CRS. With each dose of the medications, patients should be monitored for 48h for aforementioned complications.

9:

Any abnormality in renal function, including insufficiency, oliguria, and failure, increases the risk for developing tumor lysis syndrome (Cairo et al., Br J Haematol 2010, 149: 578-86).

10:

Many chemotherapies can cause acute kidney injury (AKI) and therefore require inpatient administration with close kidney function monitoring (Money et al., Front Oncol 2021, 11: 607574).

11:
Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

12:
Interleukin-2 is significantly toxic to the recipient. Those receiving this therapy must be monitored closely during and after administration (Rathmell et al., J Clin Oncol 2022, 40: 2957-95; Tallman et al., J Natl Compr Canc Netw 2019, 17: 721-49).

13:
Methotrexate administered at a dosage of greater than 0.5 g/m^2 requires close monitoring, aggressive hydration, and urinary alkalinization. Rescue doses of folinic acid or glucarpidase are typically administered 24 to 36 hours after methotrexate initiation to avoid life-threatening nephrotoxicity, and are continued until methotrexate levels fall below 0.05 to 0.1 μM/L (Brown et al., Journal of the National Comprehensive Cancer Network 2019, 17: 414-23; Tallman et al., J Natl Compr Canc Netw 2019, 17: 721-49).

14:
The most common indications for bone marrow or stem cell transplants are leukemia, lymphoma, multiple myeloma, solid tumor cancers, or certain non-malignant conditions (e.g., congenital defects). Chemotherapy, radiation, and/or immunosuppressive drugs are used to prepare the patient for the transplant.

15:
The period preceding a bone marrow or stem cell transplant is used to prepare the patient for the transplant by completely suppressing the bone marrow and the patient's immune response. This conditioning regimen can consist of either or both chemotherapy and radiotherapy (Bazinet and Popradi, Curr Oncol 2019, 26: 187-91). The American Cancer Society suggests that 1-2 weeks is an appropriate time frame for patients to be admitted prior to a bone marrow or stem cell transplant. InterQual® external peer reviewers agree that 10 days captures the majority of the patients admitted for this treatment.

16:
Prior to receiving CAR T-cell therapy, patients undergo a lymphodepletion regimen. This regimen requires frequent lab and vital sign monitoring to ensure their immune system is being suppressed. The goal of lymphodepletion is to increase the effectiveness of the CAR T-cell treatment (Thompson et al., J Natl Compr Canc Netw 2022, 20: 387-405; Santomasso et al., J Clin Oncol 2021, 39: 3978-92).

17:
Iodine ablation is radiation therapy used to ablate or destroy remaining thyroid tissue after a thyroidectomy is performed (James et al., JAMA Otolaryngol Head Neck Surg 2021, 147: 544-52; Piccardo et al., J Nucl Med 2020, 61: 1730-5).

18:
Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

19:
High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less

oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

20:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

21:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

22:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(*0.21*)

1L 24%(*0.24*) **4L** 36%(*0.36*)

2L 28%(*0.28*) **5L** 40%(*0.40*)

3L 32%(*0.32*) **6L** 44%(*0.44*)

Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

23:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

24:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.
These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

25:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).
Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

26:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

27:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

28:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

29:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

30:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

31:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

32:

Cytokine release syndrome can occur after immunotherapies such as CAR (chimeric antigen receptor) T-cell therapy and it can also occur after bone marrow or stem cell transplants. It is characterized by a fever and a systematic inflammatory response (Thompson et al., J Natl Compr Canc Netw 2022, 20: 387-405; Abboud et al., Bone Marrow Transplant 2021, 56: 2763-70; Santomasso et al., J Clin Oncol 2021, 39: 3978-92).

33:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.
AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.
AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

34:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

35:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

36:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J

Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.
Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

37:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

38:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

39:

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

40:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

41:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

42:

Tumor lysis syndrome can be a serious complication of cancer treatments. It is characterized by the release of intracellular contents into the blood vessels causing metabolic abnormalities. If untreated, it can lead to renal failure, arrhythmias, seizures, loss of muscle control, or death (Alqurashi et al., Cureus 2022, 14: e30652; Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

43:

It is recommended that tumor lysis syndrome prophylaxis be initiated prior to chemotherapy. However, rasburicase is considered the first line agent in the treatment of tumor lysis syndrome. Allopurinol is an alternative therapy when rasburicase is contraindicated due to hypersensitivity reaction or G6PD deficiency (Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

44:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

45:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when

the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.
The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

46:

Hyperviscosity syndrome is caused by elevated immunoglobulin levels which coat red and white blood cells, causing increased blood viscosity, sludging of blood and hypoperfusion. It can be seen in diseases such as multiple myeloma, leukemia, polycythemia, and other myeloproliferative disorders. Symptoms of hyperviscosity syndrome include headache, tinnitus, stupor, coma, and uncontrolled or recurrent epistaxis. In the presence of symptoms, therapeutic plasma exchange is indicated (Halaseh et al., QJM 2022, 115: 30-1; Emerson and Kaide, Emerg Med Clin North Am 2018, 36: 603-8; Gertz, Blood 2018, 132: 1379-85).

47:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

48:

Repeated intravenous fluid boluses may be used to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as, lactated ringers (Myburgh, N Engl J Med 2018, 378: 862-3).
In adults, bolus volumes may be generally defined as fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34).
For patients with decreased cardiac reserve, significant renal dysfunction or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206).

49:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)

- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

50:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

51:

Dexamethasone is effective in relieving chemotherapy-induced nausea and vomiting. Patients receiving chemotherapy with a high or moderate emetic risk may be given a combination of dexamethasone, serotonin antagonists, and/or aprepitant (Olver et al., Support Care Cancer 2017, 25: 297-301).

52:

Diarrhea is one of the most common adverse effects of cancer and cancer treatment and is a frequent symptom of colitis. Diarrhea is often made worse by preexisting inflammatory bowel disease. Per the National Comprehensive Cancer Network (NCCN) Guidelines, patients experiencing diarrhea related to cancer treatments should be treated with steroids, immunosuppressants, and/or monoclonal antibody therapies (Ajani et al., J Natl Compr Canc Netw 2022, 20: 167-92; Bellaguarda and Hanauer, Am J Gastroenterol 2020, 115: 202-10; Gong and Wang, JCO Oncol Pract 2020, 16: 453-61; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

53:

Abdominal computed tomography (CT) is generally performed when patients with severe diarrhea present with abdominal pain to diagnose colitis or to rule out other intra-abdominal processes (Tang et al., Front Immunol 2021, 12: 800879; Gong and Wang, JCO Oncol Pract 2020, 16: 453-61; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

54:

Corticosteroids are the first line treatment for most inhibitor-related toxicities secondary to immunotherapy (Ben-Horin et al., Clin Gastroenterol Hepatol 2022, 20: 2868-75 e1; Fan and Naidoo, JCO Oncol Pract 2020, 16: 462-3; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

55:

Immunosuppressant therapies to treat diarrhea related to cancer or cancer treatments can include but are not limited to (Fan and Naidoo, JCO Oncol Pract 2020, 16: 462-3; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913):

- Infliximab
- Tofacitinib
- Ustekinumab
- Vedolizumab

56:

Pain management is an essential component and effective therapy for patients with cancer (Li et al., Cochrane Database Syst Rev 2021, 8: CD011108; Swarm et al., J Natl Compr Canc Netw 2019, 17: 977-1007; WHO Guidelines for the Pharmacological and Radiotherapeutic Management of Cancer Pain in Adults and Adolescents. 2018).

57:

Jaundice can be a sign of advanced disease or another complication and therefore requires a thorough workup at the Acute level of care (Kastelijn et al., J Gastrointest Cancer 2022:; An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Lee et al., HPB (Oxford) 2018, 20: 477-86).

58:

Instruction: For known Von Willebrand disease, use the Anemia/Bleeding subset.

59:

Thrombocytopenia is a frequent complication of cancer and cancer treatments. The most common treatments for thrombocytopenia are blood product transfusions, corticosteroids and immunoglobulins (Omar et al., Front Immunol 2020, 11: 1354; Schiffer et al., J Clin Oncol 2018, 36: 283-99).

60:

Intravenous immunoglobulin (IVIG) is a blood product which contains immunoglobulins extracted, then pooled, from the plasma of thousands of blood donors. Its administration carries the same risks as those of any blood product transfusion, including headache, backache, chills, nausea, muscle pain, and, although rare, anaphylaxis. IVIG is used for a wide range of disorders, and its therapeutic mechanism varies based on the dose given and the patient's disease. IVIG is used for the treatment of diseases in neurology, hematology, immunology, dermatology, and nephrology.

61:

Aggressive hydration is an essential component of prevention and treatment of tumor lysis syndrome (Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

62:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

63:

Instruction: This line of criteria may be appropriate for patients who received their bone marrow or stem cell transplant on Episode Day 1. It is also appropriate for patients who received myeloablative, sub-myeloablative or lymphodepletion therapy in the outpatient setting and were admitted one day before their bone marrow or stem cell transplant.

64:

Grades 1 and 2 of cytokine release syndrome (CRS) are usually characterized by a fever alone or fever with hypoxia and/or hypotension, respectively. Interventions for CRS can include broad spectrum corticosteroids, IV fluids, and anticytokine therapies. If fever persists, tocilizumab which is usually given to Grade 2 CRS can be considered (Thompson et al., J Natl Compr Canc Netw 2022, 20: 387-405; Santomasso et al., J Clin Oncol 2021, 39: 3978-92; Lee et al., Biol Blood Marrow Transplant 2019, 25: 625-38).

65:

Aphasia for patients with neurotoxicity can include but is not limited to episodes of expressive, global, or receptive aphasia.

66:

Hyperkalemia (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50; Depret et al., Ann Intensive Care 2019, 9: 32; Montford and Linas, J Am Soc Nephrol 2017, 28: 3155-65).

67:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

68:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

69:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

70:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

71:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

72:

Blood pressure is considered labile when it is extremely variable and difficult to control.

73:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

74:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

75:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

76:

Graft versus Host Disease (GVHD) is a transplant complication that occurs when immunocompetent cells from a non-identical donor are grafted into an immunosuppressed recipient. The immunocompetent donor/graft cells recognize alloantigens in the host cells and the host develops a Graft vs Host (GVH) reaction. Classic acute GVHD typically presents during the first 100 days after a hematopoietic stem cell transplantation (HSCT). Classic chronic GVHD presents beyond the first 100 days post HSCT and may include overlap syndrome, with acute GVHD features. Classic acute GVHD affects the skin, gut, and liver. Diagnosis is strongly supported if symptoms include exanthema (an eruptive skin rash oftentimes associated with a fever or systemic symptoms), diarrhea, and elevated bilirubin levels. Chronic GVHD, in addition to skin, liver, or gut involvement, may also affect the mouth, eyes, respiratory tract, musculoskeletal system, and peripheral nerves. It presents with symptomatology resembling autoimmune diseases (e.g., systemic lupus, erythematosus, or rheumatoid arthritis).
Chronic GVHD grade can be categorized as mild, moderate, or severe. In patients with GVHD, hyperbilirubinemia and small intestinal/colonic involvement are known risk factors for increased mortality (Hill et al., Annu Rev Immunol 2021, 39: 19-49; Penack et al., Lancet Haematol 2020, 7: e157-e67; Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50).
In determining GVHD grade, stages for skin, liver and gut are scored.
Acute GVHD Staging and Grading (Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50):
Grade I: Stage 1-2 skin organ involvement, only
Grade II: Stage 3 skin; Stage 1 liver and gut
Grade III: Stage 1-3 skin; Stage 2-3 liver and Stage 2-4 gut
Grade IV: Stage 4 skin and liver
Stage 1

- Skin: Rash < 25%(0.25) of body surface area
- Liver: Bilirubin 2-3 mg/dL (35-40 µmol/L)
- Gut: Diarrhea > 500-1000 ml/day

Stage 2

- Skin: Rash 25%(0.25) to 50%(0.50) of body surface area with associated symptoms
- Liver: Bilirubin 3-6 mg/dL (53-102 µmol/L)
- Gut: Diarrhea > 1000 ml/day

Stage 3

- Skin: Generalized rash covering > 50%(0.50) of body surface area
- Liver: Bilirubin 6-15 mg/dL (104-225 µmol/L)
- Gut: Diarrhea > 1500 ml/day

Stage 4

- Skin: Generalized erythroderma (reddening of skin often associated with exfoliation) or bullous eruption
- Liver: Bilirubin > 15 mg/dL (225 µmol/L)
- Gut: Severe abdominal pain with or without ileus

Chronic GVHD grading (scores are based on a 4-point scale of 0-3):

- Mild: 1 or 2 organs, with score no more than 1 with lung score 0
- Moderate: 3 or more organs with no more than score 1 or > 1 organ (not lung) with score 2 or lung score 1
- Severe: At least 1 organ with score of 3 or lung score of 2 or 3

77:
Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

78:
Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

79:
Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.
Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

80:
Calcium levels within criteria represent total serum calcium.

81:
Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

82:
Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

83:
Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to

ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

84:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

85:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

86:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

87:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm(500×10^6/L) are severely neutropenic and have a very limited immunologic response to infection.
To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.
Examples:
ANC calculation using a percentage of total WBC count:
WBC count: 1,500/cu.mm($1.5x10^9$/L)
Segs: 20%(0.20) of the WBCs
Bands: 2%(0.02) of the WBCs
(20% + 2%(0.20 + 0.02)) x 1,500/cu.mm($1.5x10^9$/L)
= 330/cu.mm($330x10^6$/L)
Normal range: 1,500-8,000/cu.mm($1.5\text{-}8.0x10^9$/L)
An ANC of less than 500/cu.mm($500x10^6$/L) is considered severely neutropenic.
ANC Calculation using the above numbers as whole numbers:
WBC count: 1.5
Segs: 0.2
Bands: 0.02
(0.2 + 0.02) x 1.5 = 0.33
Normal range: 1.5 to 8.0
An ANC of less than 0.5 is considered severely neutropenic.

88:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

89:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

90:

Graft dysfunction is usually treated empirically with anti-rejection therapy while awaiting the results of the biopsy. Immunosuppressive medications are initiated based on the signs and symptoms of organ dysfunction or graft versus host disease (GVHD).

91:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

92:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

93:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

94:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

95:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

96:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

97:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

98:

Hyperviscosity syndrome may be a complication of a hematologic malignancy and is characterized by an extreme elevation of hematocrit, immature white blood cells (WBC), or paraprotein (IgM, IgG, IgA). Symptoms may include neurologic deficits, acute renal failure, and cardiac ischemia.

99:

Recommendations regarding evaluation, treatment, and setting of care for patients with fever and neutropenia are based on risk stratification into low and high risk groups. Assessment of the risk for complications from a severe infection should be performed upon presentation.
Low risk patients are generally defined by clinical stability without an active medical comorbidity (e.g., mucositis, nausea and vomiting, mental status change, etc.). High risk factors may include, but are not limited to, the following:

- Absolute neutrophil count (ANC) ≤ 100/cu.mm($100x10^6$/L)
- Mental status change
- Chronic lung disease
- Abdominal pain, new onset
- Mucositis
- Nausea and vomiting
- Diarrhea
- Uncontrolled pain
- Underlying malignancy with bone marrow involvement (e.g., acute leukemia)

In addition, use of validated risk assessment tools which take into account a multitude of risk factors, such as the Multinational Association for Supportive Care in Cancer (MASCC) scoring system, are recommended to categorize patients into low and high risk groups. Pediatric guidelines specifically recommend that hospitals adopt and use a validated risk stratification tool to guide routine management. Select low risk patients may be candidates for outpatient management, but inpatient management remains the standard of care (Pettit et al., Am J Emerg Med 2021, 50: 5-9; Zimmer and Freifeld, J Oncol Pract 2019, 15: 19-24).

100:

Febrile neutropenia is a medical emergency for oncology and hematology patients. The Multinational Association for Supportive Care in Cancer (MASCC) Index is the recommended tool for the risk stratification of these patients (Pettit et al., Am J Emerg Med 2021, 50: 5-9).

101:

An absolute neutrophil count that is expected decline to < 500/cu.mm($500x10^6$/L) within 48 hours is included in the definition of neutropenia (National Comprehensive Cancer Network, The NCCN clinical practice guidelines in oncology, Prevention and Treatment of Cancer-Related Infections (Version 2.2023). 2023).

102:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

103:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).
Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

104:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

105:

All medications commonly used to prevent rejection or opportunistic infections in the post-transplant period cause some degree of side effects. One of the important tasks of the transplant team is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is attempted. Tacrolimus and cyclosporine (CSA) are immunosuppressive agents that inhibit calcineurin and are given to solid organ transplant recipients to prevent and treat allograft rejection. Tacrolimus is less nephrotoxic and may be more cost-effective than CSA. However, new-onset diabetes should be closely monitored during the medication period. The dosing of CSA is more difficult to adjust appropriately in children, especially in younger/smaller children (less than 2
years). This is thought to be due to their shorter intestine, leading to less absorption of the drug, and also their faster metabolism. More time may be needed to adjust the dose and determine the appropriate medication management in this population than in older children or adults. The goal is to preserve the graft without seriously harming the patient.

106:

Patients who develop bleeding while receiving anticoagulation should be evaluated for an underlying source of bleeding. Evaluation may include upper and lower gastrointestinal (GI) studies to rule out underlying GI pathology or an imaging study to rule out intracranial bleeding (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3).

107:

Orthostatic hypotension is a common complication for patients after chemotherapy or immunotherapy due to lack of oral intake and progressive dehydration. These patients frequently require close monitoring and intravenous fluids (Vecchie et al., Cardiooncology 2021, 7: 40).

108:

Instruction: In some institutions, the first 3 doses of muromonab-CD3 are required to be given in Intermediate care when the patient is experiencing acute rejection. The initial doses of muromonab-CD3 can cause violent headaches, fevers, and dyspnea. Dyspnea is more common in fluid overloaded patients and can resemble acute respiratory distress syndrome (ARDS) in severe forms.

109:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

110:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

111:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

112:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

113:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

114:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

115:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

116:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

117:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

118:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

119:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

120:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

121:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

122:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work

environment)

- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

123:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

124:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

125:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

126:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

127:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

128:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

129:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.